

MANOBHI SRI RAM

Generative AI Engineer · Agentic AI Systems · LLM Orchestration · RAG Pipelines · LLMOps
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PROFESSIONAL SUMMARY

Generative AI Engineer building production multi-agent systems across enterprise operations, legal, and healthcare domains — from a Bedrock-orchestrated travel operations platform adopted by 5,000+ employees across 12 countries, to a legal-AI pipeline cutting contract review from hours to under a minute at \$0.05/review. Built with production discipline from day one: safety guardrails, human-in-the-loop escalation, adversarial red-teaming, and full LLMOps observability across every system shipped.

WORK EXPERIENCE

ASCENDION | AI Engineer — Generative AI & ML · Chennai, TN

Aug 2025 – Present

Leadership Mobility Command Center — Generative AI + Agentic AI Enterprise Operations

Core AI Lead | Led a 5-member team

Deployed across 12 countries for 5,000+ employees — automated end-to-end trip coordination, saving 300+ coordinator hours/month at \$0.08–\$0.19 inference cost per trip.

- Built **LangGraph multi-agent orchestration** (flight + ground transport agents, native state checkpointing, retry/fallback on agent failure or Bedrock API downtime/rate-limits) with model routing via **AWS Bedrock** — Claude Haiku for lightweight tasks, Claude Sonnet 4.5 for complex agentic reasoning — keeping per-trip inference cost at **\$0.08–\$0.19** while maintaining decision quality.
- Implemented **AWS Bedrock Guardrails** for content safety filtering, blocking unsafe or off-topic agent outputs and escalating flagged cases to a human operator; designed **human-in-the-loop approval** via Teams for high-stakes actions (rebooking, cancellations), with edge cases (visa, weather, VIP prefs) also escalated to humans.
- Saved **300+ coordinator hours/month** by delivering a production RAG pipeline (LangChain, AWS Bedrock Titan Embeddings, Qdrant) over **13,000+ vectorized trip records** — record-level chunking and hybrid keyword+vector retrieval eliminated manual coordination for standard cases.
- Validated **near-100% agent decision accuracy** against real trip scenarios before enabling autonomy, then deployed on **AWS EC2 with CI/CD** and self-managed Prometheus + Grafana monitoring for automated rollback on failure.
- Integrated Aviationstack delay forecasting, Google Maps, Microsoft Teams, and Twilio/Firebase GPS for real-time trip intelligence, with **PII masking** to meet multi-country data compliance requirements.

Legal Contract Intelligence Platform — Agentic AI · Multi-Model · MCP · LLMOps

Core AI Lead | Led a 5-member team

Reviews contracts in under a minute for ₹4.48 (\$0.054) vs 2–4 hrs of manual review — saves ~₹1,400 (\$16.87) per review. Validated across 150+ real contracts spanning multiple contract types.

- Architected a **4-agent AutoGen pipeline** — chosen over LangGraph for true agent-to-agent debate — where Claude Sonnet 4.5 extracts clauses, GPT-4o classifies risk against 47 known patterns, Gemini 1.5 Pro flags deviations from standard templates, and Claude Opus 4.5 synthesizes deal-breakers and negotiation priorities into the final report.
- Extended the pipeline with Claude/GPT-4o Vision to pull tables, stamps, and handwritten redlines directly from scanned contract pages.
- Delivered a RAG search system (Milvus, OpenAI embeddings, Cohere reranking) for plain-English queries across the contract library — **34.2% cache hit rate** cut repeat-query cost to near-zero — plus a scheduled agent that checks 5 regulatory sources daily and auto-flags contracts impacted by new laws.
- Exposed the entire system as **4 MCP servers (17 tools)** so it runs natively inside Claude Desktop — no custom integration code, users just ask in plain English and Claude calls the right tool.
- Validated reliability pre-launch: LLM-judge evaluation scored **82.4% accuracy** across 100 CUAD contracts, adversarial red-teaming cut failure rate **from 41% to 9%** over 3 rounds, and a 46-function pytest suite with full LLM-API mocking — monitored in production via Datadog APM and Langfuse for full cost/latency/error visibility.

PROJECTS

MedAssist — Clinical Triage Agent (Healthcare AI)

2025 [GitHub](#)

Fine-tuned, locally-deployed clinical triage agent — turns plain-English symptoms into a source-cited, WHO/ICMR-grounded triage decision. Zero cloud dependency, zero cost per query.

- Fine-tuned Phi-3 Mini 3.8B on 15,000 medical QA examples via LoRA on a T4 GPU — training loss dropped **1.92→1.21** — deployed locally via Ollama for private, zero-cost inference with zero cloud dependency.

- Designed a 5-node LangGraph agent (parse symptoms → assess severity → retrieve guidelines + check drugs → decide) with conditional routing that skips unnecessary drug-check steps, cutting latency ~30%; every output is Pydantic-validated with automatic retry to prevent malformed decisions.
- Built hybrid RAG on LanceDB (vector + full-text, custom reciprocal rank fusion) over 65 clinical documents with live OpenFDA drug-contraindication lookups — semantic chunking raised context precision **from 0.65 to 0.85** — with every decision fully traced via Langfuse for explainable, auditable reasoning.
- Built with **responsible AI principles**: a clinical disclaimer positions the system as a triage aid, not a diagnostic replacement, and outputs were validated with Ragas (0.92 faithfulness, 0.90 relevancy, 0.85 precision, 0.87 recall) with GitHub Actions CI/CD blocking any push that drops faithfulness below 0.60.

TECHNICAL SKILLS

Agentic AI and LLMs: LangChain, LangGraph, AutoGen, CrewAI, MCP Server Development, Claude (Anthropic), GPT-4o, Gemini, Groq, Prompt Engineering, LLM Fine-Tuning (LoRA/QLoRA)

RAG and Vector Search: Retrieval Augmented Generation (RAG), Pinecone, Qdrant, Weaviate, Milvus, LanceDB, FAISS, Cohere Rerank, Hybrid Search (BM25 + Vector), Semantic Chunking, RRF Fusion, Vector Embeddings

LLMOps and Model Evaluation: Langfuse, Datadog, Prometheus, Grafana, Ragas, DeepEval, Hugging Face, MLOps

Machine Learning and AI: Deep Learning, Natural Language Processing (NLP), Computer Vision, Ollama, AI Agents

Backend and Cloud: FastAPI, REST API, PostgreSQL, Docker, AWS (EC2, S3, Bedrock, CloudWatch, SageMaker), Azure, Git, GitHub, Render, SQLAlchemy

Languages and Frontend: Python, SQL, JavaScript, React, Next.js, Streamlit

CERTIFICATIONS

- **Microsoft Certified: Azure AI Developer Associate** · Microsoft · 2026
- **AWS Certified Cloud Practitioner** · Amazon Web Services · 2023
- **AWS Certified AI Practitioner** · Amazon Web Services · 2026

EDUCATION

Vellore Institute of Technology (VIT), Chennai | B.Tech, Computer Science & Engineering

Aug 2021 – Aug 2025

CGPA: 8.39 / 10

B.Tech Final Year Thesis — AI Based Post-Harvest Management of Mushrooms · 2025

Best Project Award · First Author

- Built a custom 2,000+ image dataset from scratch and trained a CNN (**R² 0.92, MSE 7.8, MAE 1.9**) — outperformed manual inspection (65% accuracy) and traditional image processing (75% accuracy) in a comparative evaluation, with data augmentation lifting accuracy **from 0.84 to 0.92 R²**.